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Radiomics for differentiating adenocarcinoma and squamous cell carcinoma in non-small cell lung cancer beyond nodule morphology in chest CT

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Distinguishing between primary adenocarcinoma (AC) and squamous cell carcinoma (SCC) within non-small cell lung cancer (NSCLC) tumours holds significant management implications. We assessed the performance of radiomics-based models in distinguishing primary there is from SCC presenting as lung nodules on Computed Tomography (CT) scans. We studied individuals with histopathologically proven adenocarcinoma or SCC type NSCLC tumours, detected as lung nodules on Chest CT. The workflow comprised manual nodule segmentation, regions of interest creation, preprocessing data, feature extraction, and nodule classification using machine learning algorithms. The dataset comprised 46 adenocarcinoma and 28 SCC cases. For feature extraction, 101 radiomic features were extracted from the tumour regions using the 'pyradiomics' module in Python. After extensive experimentation with various feature importance techniques, the top 10 most significant radiomic features for differentiating between adenocarcinoma and squamous cell carcinoma (SCC) were identified. The Synthetic Minority Over-Sampling Technique was used to achieve a balanced distribution. Lung nodules were classified using 13 machine-learning algorithms, including Linear Discriminant Analysis, Random Forest, AdaBoost, and eXtreme Gradient Boosting. The Multilayer Perceptron (MLP) Classifier with Rectified Linear Unit (ReLu) activation was the most accurate (83% accuracy) with 83% precision and 86% sensitivity in distinguishing SCC from adenocarcinoma. It achieved a balanced F1 score of 83%, indicating well-rounded performance in both precision and sensitivity. The average Area Under the Curve score was 88%, representing good discrimination between the two classes of lung nodules. Radiomics is a powerful non-invasive tool that could potentially add to visual information obtained on CT. The MLP Classifier with ReLu activation showed good accuracy in distinguishing primary lung adenocarcinoma from SCC nodules. However, widespread multicentre trials are required to realize the full potential of radiomics in lung nodules.

Keywords Lung nodule, Chest CT, Radiomics, Adenocarcinoma, Squamous cell carcinoma, Lung carcinoma

Lung cancer is the leading cause of cancer-related deaths worldwide[1]. It is a heterogeneous disease with many clinically important subtypes[2]. Among these, histologic phenotype is a particularly important predictor of response to therapy and overall clinical outcome. More than 80% of all primary lung cancers are classified as non-small cell lung cancer (NSCLC). The major histological types of NSCLC include adenocarcinoma, and squamous cell carcinoma (SCC); deriving from small and large airway epithelia respectively[1,2]. Differentiating between AC and SCC is of clinical significance as Chemotherapy regimens for AC and SCC are different according to the National Comprehensive Cancer Network (NCCN) guidelines for NSCLC[3–5]. Proper identification of AC and SCC in NSCLC is also important for optimizing treatment strategies, particularly as some therapies, such as targeted drugs and immunotherapies, are more effective in one subtype over the other[6]. Moreover,

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the prognosis, disease progression, and potential complications differ between these subtypes, which impacts patient management and therapeutic strategies[7].

Currently, histopathology is considered the gold standard for histological classification of lung nodules[8]. However, biopsies can be limited by inter- and intra-tumour heterogeneity, which may result in an incomplete characterization of the tumour's morphology and phenotype[9,10]. Although molecular testing can identify specific driver mutations for precision medicine[9–11], its integration into routine diagnostics is often hampered by the need for specialized expertise and high costs. This highlights the need for more comprehensive diagnostic tools that can overcome these limitations. Furthermore, invasive biopsies carry risks such as pneumothorax, hemorrhage, pleural reactions, and even tumour seeding, particularly in high-risk patients[12]. Repeated biopsies, which may be necessary in the case of necrotic or heterogeneous tumours, add further complexity and increase patient burden[13]. Given these challenges, developing non-invasive diagnostic methods that can reliably distinguish between AC and SCC before treatment initiation is critical to improving patient outcomes and reducing procedural risks[14].

In recent years, advances in radiomics, a method that extracts a large number of quantitative imaging features from medical images, have offered a promising alternative for non-invasive tumour characterisation. Radiomics allows for a comprehensive, three-dimensional analysis of the entire tumour, which contrasts with the limitations of biopsy, and has been shown to improve diagnostic accuracy, especially in cancer research[15,16]. Few recent studies have also demonstrated that radiomics can successfully differentiate between lung adenocarcinoma and squamous cell carcinoma. These findings highlight the potential of radiomics to enhance the diagnostic process and guide treatment decisions more effectively[17,18]. In this study, we aim to explore the use of radiomics to predict the histological subtypes of NSCLC (AC vs. SCC) from standard-of-care computed tomography (CT) images. By harnessing non-invasive imaging techniques, this approach could provide a more comprehensive and safer alternative to biopsy-based diagnostics. A preliminary account of this work has been presented elsewhere[19].

Materials and methods

Data acquisition

Data Collection: A cross-sectional observational study (retrospective and prospective) was performed in our department after obtaining prior approval from the Institutional Research Ethics Committee – Sri Ramachandra Institute of Higher Education and Research (CSP–MED/19/SEP/56/122). All methods were performed according to the relevant guidelines and regulations. We studied individuals with histopathologically proven adenocarcinoma or SCC type of NSCLC tumours, detected as lung nodules on Chest CT during the period 2017–2023. We initially examined CT thorax scans of 1280 patients, identifying solid pulmonary nodules (3–30 mm) in 274 cases. Patients with subsolid pulmonary nodules were excluded from the study to compare purely solid pulmonary nodules. Of these, a final diagnosis was available in 180 cases. After excluding 12 cases that were not appropriate for radiomics analysis (they were touching the ribs or had calcification within), we analysed 168 nodules, of which 105 were malignant and 63 benign. There were 74 cases of primary lung carcinoma and 31 cases of metastases. Of the primary lung carcinomas, 46 were adenocarcinomas and 28 were squamous cell carcinomas (Fig. 1). We utilized convenient sampling, as this was a single-center study, and cases were limited by the requirement for biopsy confirmation. This led to a relatively smaller sample size of 46 adenocarcinomas and 28 squamous cell carcinomas. Subsequently, the Digital Imaging and Communications in Medicine (DICOM)

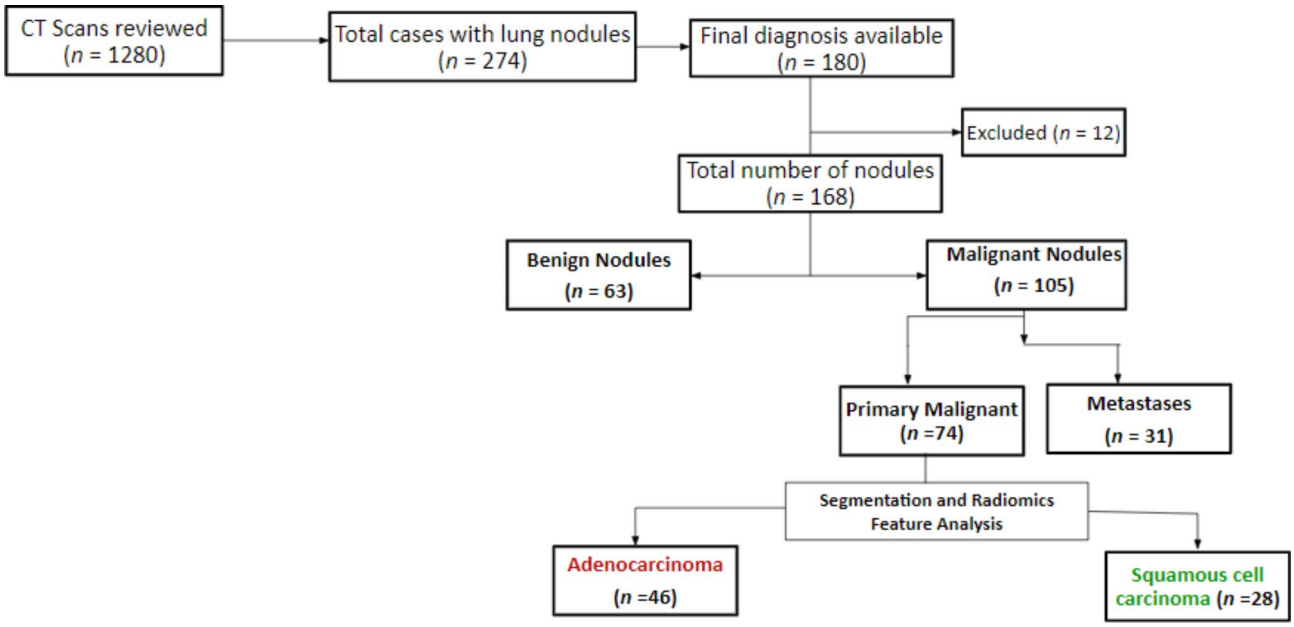


Fig. 1. Case distribution of lung nodules.

images of 46 adenocarcinoma cases and 28 squamous cell carcinoma nodules were subjected to segmentation analysis, radiomics post-processing, and feature extraction.

In our study, we specifically focused on pure adenocarcinoma and squamous cell carcinoma, intentionally excluding adenosquamous and other mixed histological types. This decision was made to ensure a clear comparison between the two primary non-small cell lung carcinoma subtypes, without introducing overlapping cases. As this is an initial study, we aimed to establish a solid foundation for future research that could explore more complex or mixed pathological types. Only patients with no other synchronous lung pathology identified were included to prevent overlap of pathologies.

Demographic Details: The age range of the study group was 18 to 83 years, with both adenocarcinoma ($n=27$) and squamous cell carcinoma nodules ($n=14$) peaking in the sixth decade. The study population exhibited a male predominance ($n=56$ males and $n=18$ females), *see* Fig. 2.

CT Acquisition: High-resolution computed tomography (HRCT) scans were conducted using one of the following multi-slice computed tomography (MDCT) scanners: the GE EVO Evolution 128-slice scanner (GE Healthcare, Princeton), the Philips Brilliance 16-slice scanner (Philips Medical Systems, Cleveland), and the PET/CT scanner Siemens Biograph Horizon (Siemens AG, Munich). HRCT images were captured during a breath-hold using 200 mA and 120 kV settings. The images were reconstructed with intervals and section thicknesses ranging from 0.65 to 0.80 mm. Subsequently, the CT images were transferred to a picture archiving and communication system (PACS) for interpretation by radiologists at dedicated workstations[20,21].

Segmentation: A radiologist segmented the DICOM images of the pulmonary nodules using the Insight Segmentation and Registration Toolkit (ITK-SNAP) software[22] with the results independently verified by three other radiologists[20,21]. Figures 3 and 4 illustrate the steps outlined above.

Workflow/methodology

The workflow comprised manual nodule segmentation, regions of interest creation, preprocessing data, feature extraction, and nodule classification using machine learning algorithms. The initial step involved manual segmentation of nodules from lung CT scans by experienced radiologists. This process ensured the accurate identification of nodule boundaries, which is important for precise feature extraction. The segmented nodule images underwent preprocessing to standardise the input for feature extraction. Utilising the 'pyradiomics' module, 101 radiomic features were extracted from the ROI of each tumour[23]. These features encompassed a wide range of characteristics, including shape, texture, and intensity-based features etc. Given the disparity in the number of adenocarcinoma (46 cases) and SCC (28 cases), the Synthetic Minority Over-sampling Technique (SMOTE)[24] was applied. This technique generated synthetic examples of the minority class (SCC) to achieve a balanced distribution, which is crucial for unbiased model training and evaluation.

Given the limited dataset and the presence of 101 features, initial classification experiments did not yield satisfactory results. To improve performance, we explored multiple methods for feature importance ranking, including Least Absolute Shrinkage and Selection Operator (LASSO)[25], Minimum Redundancy Maximum Relevance (mRMR)[26], and Random Forest Classifier[27]. We applied 5-fold cross-validation to perform hyperparameter tuning and identify the best-performing model for LASSO and Random Forest Classifier. Feature importance was then calculated for all features using each of the three methods, focusing on their ability

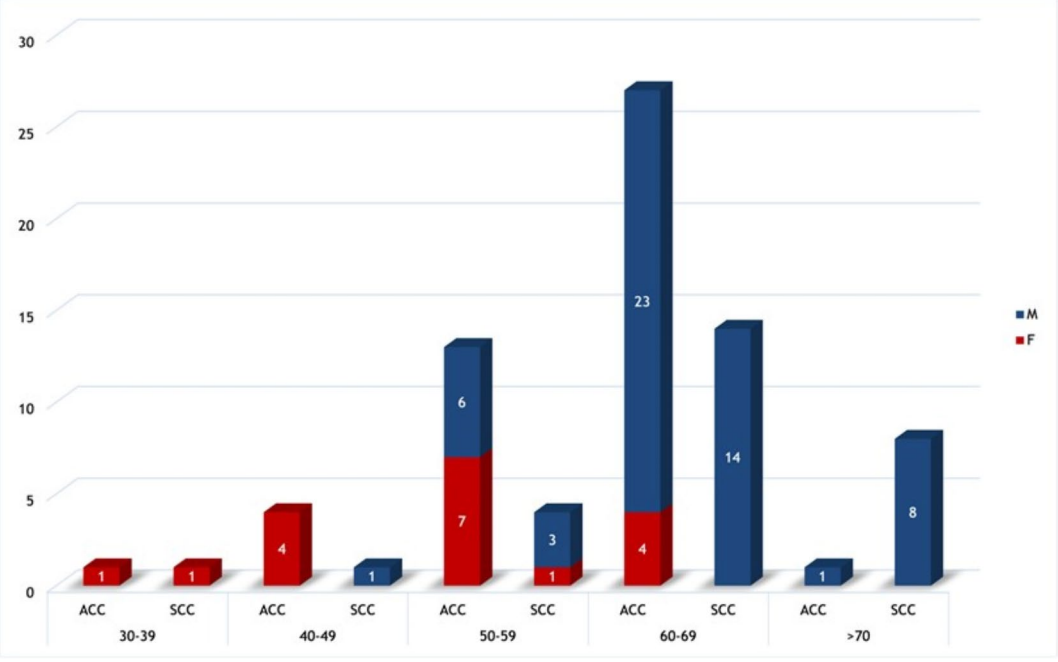


Fig. 2. Age and sex distribution of adenocarcinoma and squamous cell carcinoma lung nodules.

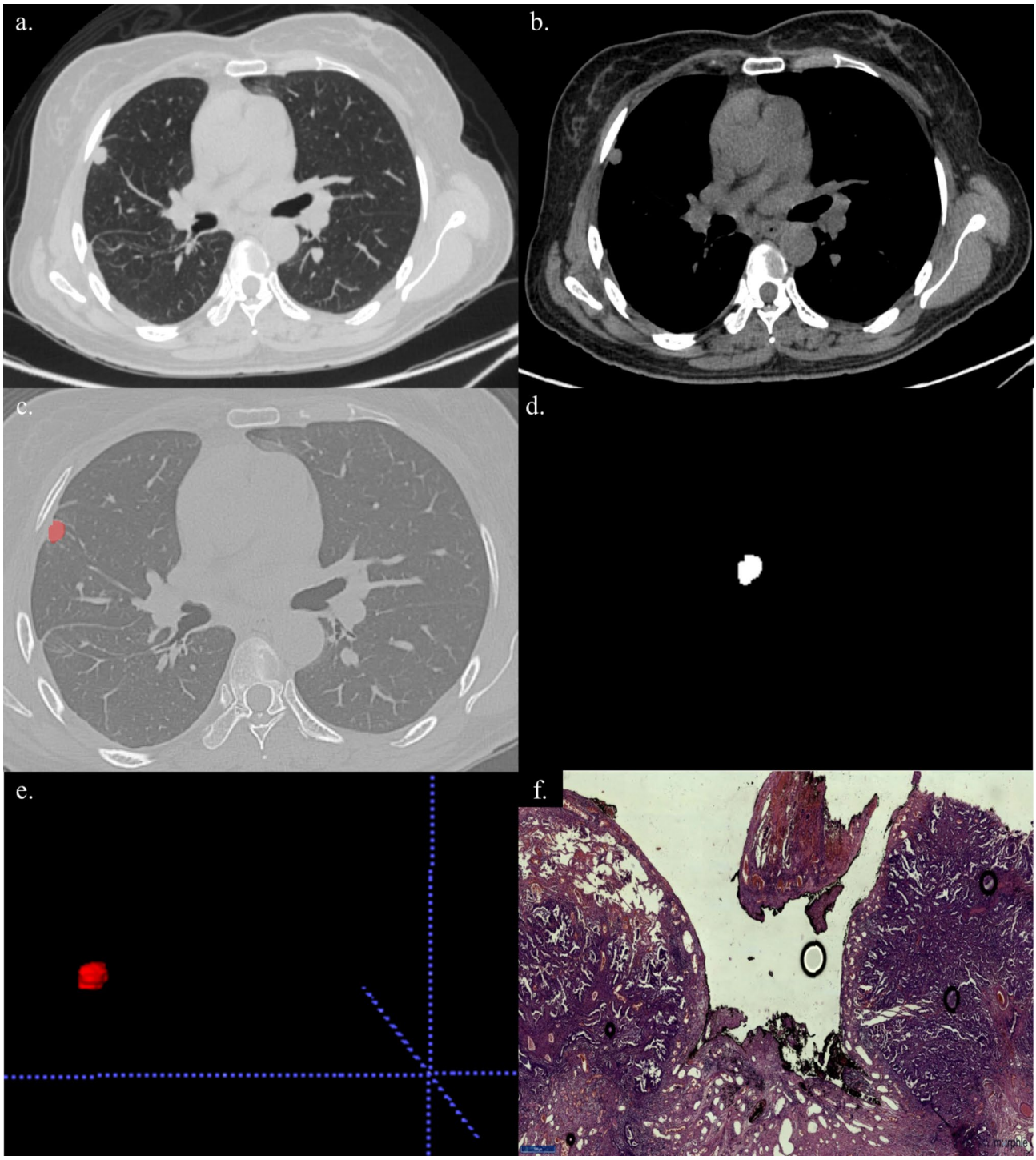


Fig. 3. A 54-year-old lady with cough. a. & b. Axial section of CT thorax in lung window and soft tissue window showing a nodule in the lateral segment of the right middle lobe. c. Creation of ROI for segmentation. d. 2D-Segmented nodule. e. 3D-Volumetric rendering of the nodule. f. HPE: Adenocarcinoma lung. Segmented nodule. e. 3D-Volumetric rendering of the nodule. f. HPE: Squamous cell carcinoma of lung.

to distinguish between adenocarcinoma and squamous cell carcinoma (SCC) provided in Table S3, S4, and S5 in the supplementary material. We experimented with different numbers of selected features and found that reducing the feature set to 10 provided the most optimal results for the classifiers provided in Figure S1, S2 in supplementary material, and Fig. 5. Among the feature selection methods, LASSO consistently produced the best classification performance, making it the primary feature selection method for our final model. LASSO aids in feature selection by applying L1 regularisation, which penalises the absolute size of the coefficients, effectively

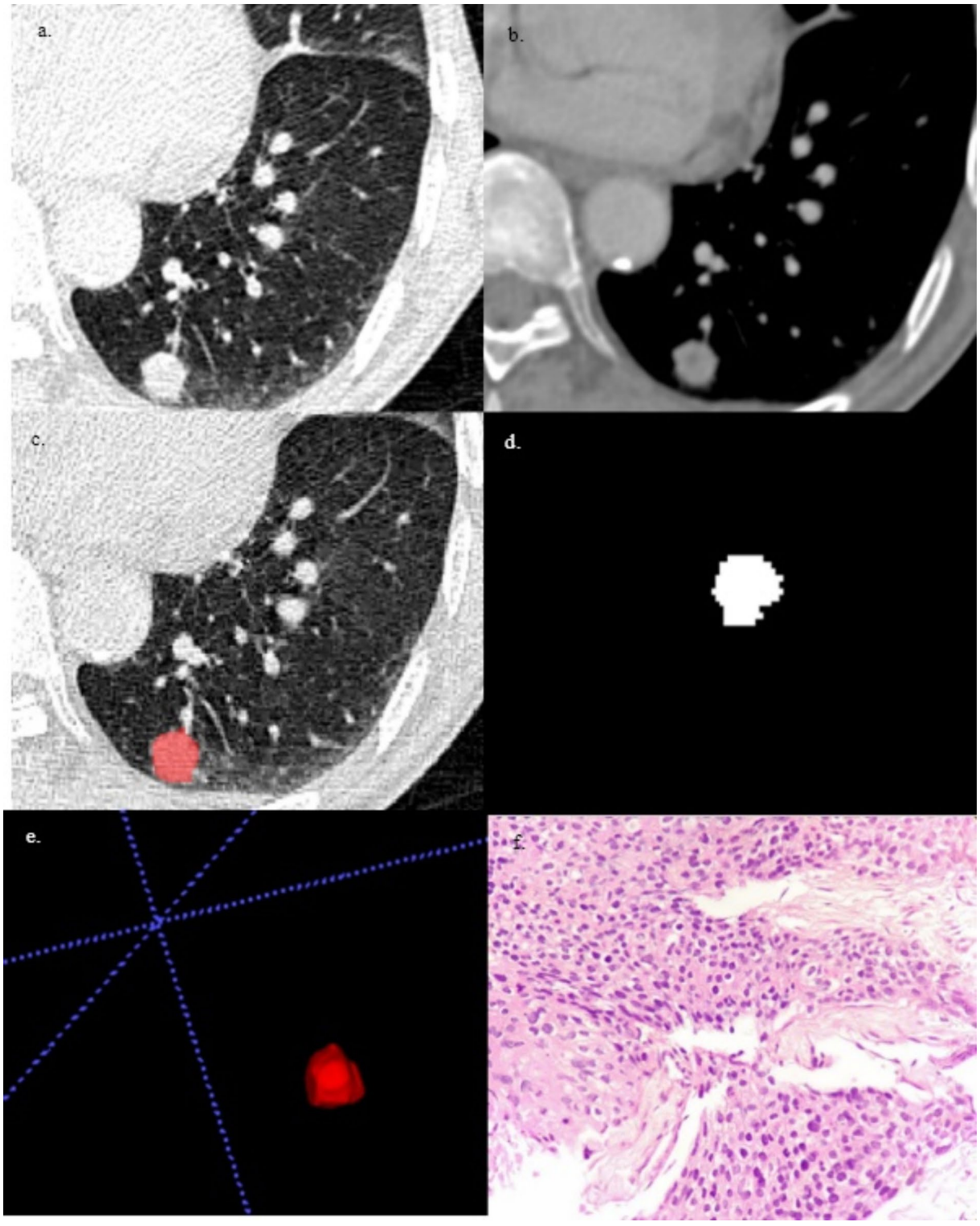


Fig. 4. A 57-year-old gentleman with right upper and lower limb weakness. MRI brain showed metastases in the brain following which CT chest was done.a. & b. Axial section of CT thorax in lung window and soft tissue window showing a nodule in the posterior basal segment of the left lower lobe.c. Creation of ROI for segmentation. d. 2D-Segmented nodule. e. 3D-Volumetric rendering of the nodule. f. HPE: Squamous cell carcinoma of lung.

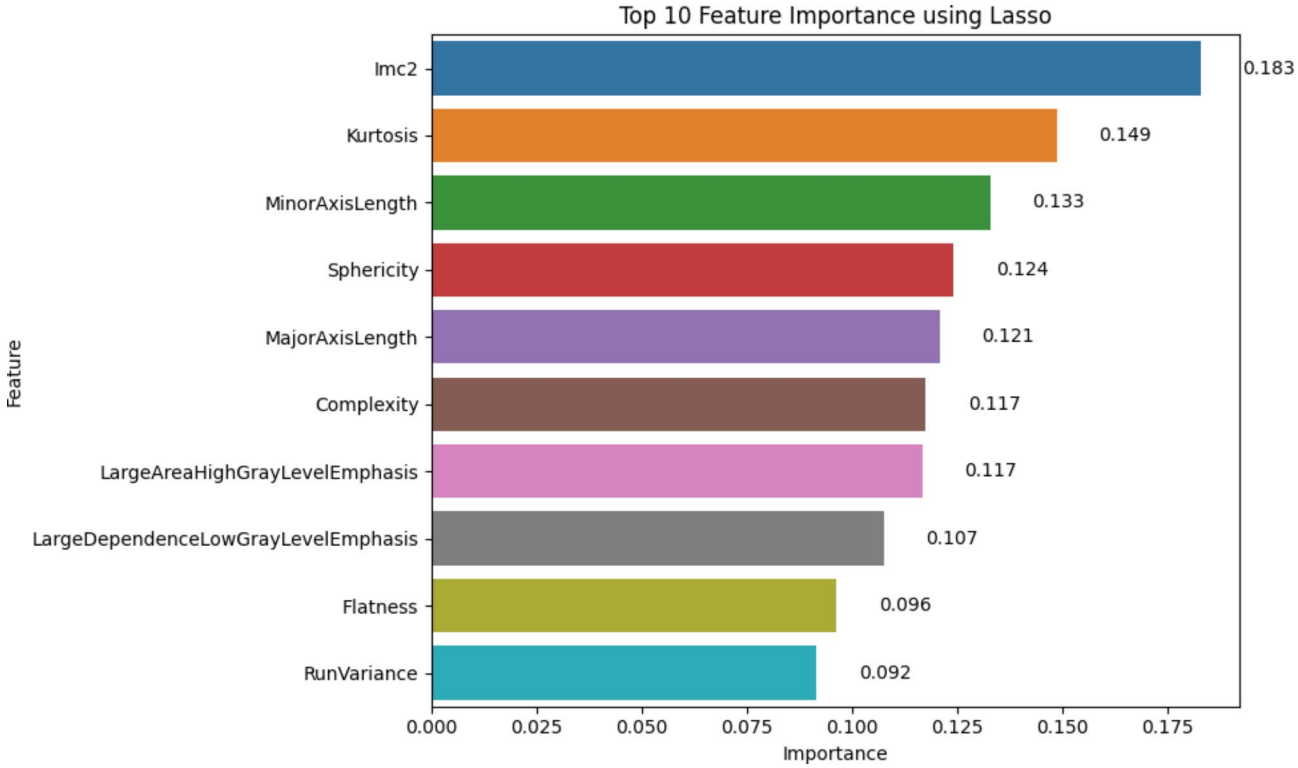


Fig. 5. Ten most important features after performing feature selection using a LASSO classifier model for the classification of Adenocarcinoma and Squamous cell carcinoma lung nodules.

driving less important feature coefficients to zero and allowing the model to select only the most relevant features for classification.

The top 10 features were selected for model training to focus on the most discriminative features, reducing the dimensionality and potentially improving model performance as shown in Fig. 5. Thirteen machine learning algorithms were trained using the selected features. These included some machine learning models along with Random Forest, AdaBoost[28], eXtreme Gradient Boosting (XGBoost)[29], and MLP classifiers. The selection of the thirteen machine learning algorithms in our study was driven by the need to evaluate a diverse range of models across different paradigms to identify which type would yield the best results for our dataset. This included tree-based algorithms (e.g., Decision Trees, Random Forests) known for their interpretability and handling of mixed data types, support vector machines (SVM) for their robustness in high-dimensional spaces, and naive Bayes and discriminant analysis for their probabilistic foundations. Additionally, multi-layer perceptrons (MLP) were included to capture non-linear relationships in the data.

Each model's performance was systematically evaluated to determine the most effective algorithm for classifying the nodules. The evaluation focused on computing the average performance metrics across all classifiers, presented with a 95% confidence interval. The summary of these performance metrics with standard deviations, is provided in Table 1 for LASSO features and Table S1 and S2 in supplementary material for mRMR and Random Forest features. Through experimentation, we found that MLP classifiers outperformed other models, particularly in capturing complex patterns. Consequently, we performed an extensive hyperparameter optimisation on the MLP models, fine-tuning aspects such as architecture, activation functions, and learning rates to further improve their performance. The hyperparameter tuning results are given in the supplementary Table S6.

Performance Metrics: The classifiers' performance was assessed using confusion matrices, from which the following metrics were derived:[20,21].

- Accuracy: The proportion of true results among the total number of cases examined.
- Sensitivity: The ability of the model to identify true positives.
- Specificity: The ability of the model to identify true negatives.
- Precision: The proportion of true positives among the total predicted positives.
- F1-Score: It is a measure of model performance that is the harmonic mean of precision and recall. It represents a trade-off between the two scores making it sensitive to both false positives and false negatives.

To ensure the study was adequately powered to detect differences between adenocarcinoma and squamous cell carcinoma, we conducted a power analysis using the independent two-sample t-test. We used the TTestIndPower function from the statsmodels library to calculate the statistical power across different effect sizes (0.2, 0.5, and 0.8), representing small, medium, and large effects, respectively. The sample sizes were 46 for adenocarcinoma

Classifiers	Accuracy	Sensitivity	Specificity	Precision	F1 score
Nearest Neighbors	0.76 ± 0.02	0.79 ± 0.03	0.73 ± 0.03	0.75 ± 0.02	0.76 ± 0.02
Linear SVM	0.71 ± 0.02	0.77 ± 0.04	0.67 ± 0.04	0.71 ± 0.03	0.71 ± 0.02
RBF SVM	0.66 ± 0.03	0.57 ± 0.06	0.79 ± 0.07	0.58 ± 0.03	0.59 ± 0.04
Decision Tree	0.69 ± 0.01	0.71 ± 0.03	0.69 ± 0.03	0.69 ± 0.03	0.69 ± 0.02
Random Forest	0.68 ± 0.02	0.68 ± 0.03	0.68 ± 0.03	0.68 ± 0.03	0.66 ± 0.02
AdaBoost	0.68 ± 0.02	0.68 ± 0.03	0.68 ± 0.03	0.69 ± 0.03	0.67 ± 0.02
Naive Bayes	0.64 ± 0.02	0.58 ± 0.05	0.72 ± 0.03	0.67 ± 0.04	0.58 ± 0.03
LDA	0.81 ± 0.02	0.85 ± 0.02	0.77 ± 0.03	0.79 ± 0.02	0.81 ± 0.02
QDA	0.75 ± 0.02	0.67 ± 0.03	0.83 ± 0.03	0.81 ± 0.03	0.71 ± 0.02
MLP Classifier_Relu	**0.83 ± 0.01**	**0.86 ± 0.03**	**0.81 ± 0.03**	**0.83 ± 0.02**	**0.83 ± 0.02**
MLP Classifier_identity	0.8 ± 0.02	0.8 ± 0.03	0.81 ± 0.02	0.81 ± 0.02	0.79 ± 0.01
MLP Classifier_tanh	0.83 ± 0.02	0.86 ± 0.03	0.82 ± 0.02	0.82 ± 0.02	0.83 ± 0.02

Table 1. The performance of various classifiers on LASSO features for distinguishing between adenocarcinoma and squamous cell carcinoma lung nodules was evaluated on the test dataset. The results are presented with 95% confidence intervals, and the best classifier is highlighted in bold. *SVM - Support vector machine; LDA - Linear Discriminant Analysis; QDA - Quadratic discriminant analysis; MLP - Multilayer perceptron.*

and 28 for squamous cell carcinoma, with a significance level (alpha) set at 0.05. The ratio of group sizes was calculated as 0.61. The power for each effect size was computed to assess the study's ability to detect meaningful differences between the two groups.

Results

Among LASSO, mRMR, and Random Forest Classifier, the LASSO method proved to be the most effective in selecting the top 10 most relevant features for distinguishing between SCC and adenocarcinoma. These features were then used to train different machine learning models, and the performance metrics were computed for each model. The best results were obtained from MLP Classifier with ReLu activation, hidden layer (100), and learning rate (0.001) stood out as the most accurate, achieving a 83% average accuracy rate in distinguishing between SCC and adenocarcinoma. This high accuracy was backed by an average precision of 83%, showing that a significant proportion of identifications were correct. With an average sensitivity of 86%, the model was relatively effective in spotting both SCC and adenocarcinoma, though there is still room for improvement. An average F1 score of 83% indicates a balance between precision and sensitivity, showcasing its well-rounded performance. Additionally, the average Area Under the Curve (AUC) score of 88% demonstrates a good ability to differentiate between the two classes, although there might be potential for further enhancement with more data, as illustrated in Figs. 6 and 7. Although the MLP model exhibited the best overall performance, it is important to note that comparative analyses with other algorithms indicate the possibility of further enhancements given the availability of more data.

We performed a decision curve analysis to evaluate the clinical utility of our predictive model across various threshold probabilities, as shown in the Fig. 8. The decision curve indicates that the model offers a notable net benefit across a wide range of threshold probabilities, particularly in comparison to the "Treat All" and "Treat None" strategies. The model demonstrates its highest net benefit from a threshold of 0.0 up to approximately 0.3, where it significantly outperforms the alternative strategies. This suggests that the model is highly effective at identifying true positives while minimizing unnecessary treatments in this lower threshold range. As the threshold probability increases beyond 0.3, the net benefit gradually decreases, with a sharp decline occurring around a threshold of 0.8. At this point, the model's net benefit drops below the "Treat None" strategy, indicating that the model may lead to missed true positives or unnecessary harm at higher thresholds. Overall, the model exhibits clinical utility primarily within the low to moderate threshold range (0.0 to 0.8), making it particularly valuable in scenarios where reducing false positives and identifying true positives are critical. Additionally, power analysis revealed that for a small effect size of 0.2, the study achieved a power of 0.13, indicating insufficient power to detect small differences between adenocarcinoma and squamous cell carcinoma. For a medium effect size of 0.5, the power was 0.54, suggesting moderate ability to detect medium-sized effects. Finally, for a large effect size of 0.8, the power was 0.91, demonstrating a strong capability to identify large effects. These results indicate that while the study is well-powered to detect large effects, it may not reliably identify small differences without a larger sample size.

Discussion

Our study highlights the significant potential of radiomics in enhancing the interpretation of malignant lung nodules, particularly in primary non-small cell lung cancer (NSCLC). By providing additional information beyond traditional visual assessment, radiomics could be integrated into Picture Archiving and Communication Systems (PACS) to improve diagnostic interpretation. Radiomics offers a non-invasive approach to augmenting the visual data obtained from CT scans. In our study, multiple methods were tested to perform feature selection such as Random Forest, LASSO, and mRMR. The figures for the Random Forest and mRMR methods are included in the Supplementary Material (Figures S1, S2). Figure 5 illustrates the feature importance derived

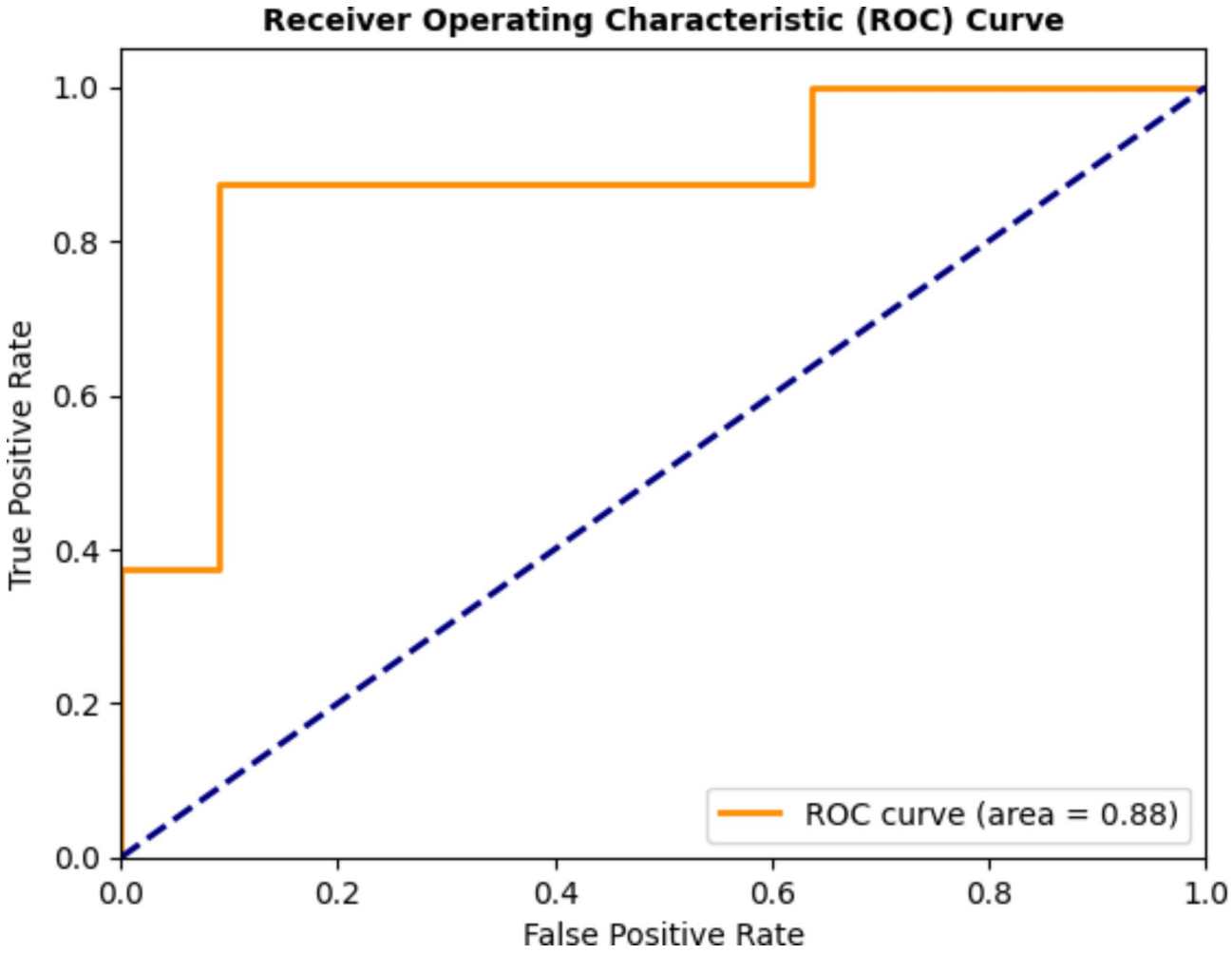


Fig. 6. Receiver Operating Characteristic (ROC) curve for the final classifier.

from the LASSO classifier, as these selected features yielded the best model performance. Once the MLP-based neural network is trained, we once again calculate the feature importance using SHapley Additive exPlanations (SHAP) as shown in Fig. 9[30]. The ranking of feature importance does change, indicating that the different models weigh the importance of the features differently to maximise the classification scores. This could also indicate that while certain models do rank features differently, most of the selected features are necessary and important towards making a good prediction for whether a given nodule represents an adeno or squamous cell carcinoma. Also, some clinical studies show that compared to squamous cell carcinoma, adenocarcinomas are more likely to have deep lobulated margins and are less likely to have smooth margins[31]. This underscores the significance of shape in differentiating between these cancer types. The Shapley features identify Sphericity as a key feature distinguishing between adenocarcinoma and SCC. This adds interpretability and clinical relevance to the model, as it aligns with observed radiological characteristics, thereby improving its practical utility in clinical settings. In our investigation, we observed that the MLP classifier with ReLU activation demonstrated a commendable accuracy rate of 83% in distinguishing between primary lung adenocarcinoma and squamous cell carcinoma (SCC) nodules. Our experimentation showed that MLP models outperformed other classifiers in capturing complex patterns, particularly after extensive hyperparameter optimisation, which involved fine-tuning the architecture, activation functions, and learning rates.

Several recent studies have demonstrated the potential of radiomics and machine learning in differentiating lung adenocarcinoma from squamous cell carcinoma[14,17,18]. In our study, we achieved an accuracy of 83% and AUC score of 88% using a radiomics-based model, with the Multilayer Perceptron (MLP) Classifier and Rectified Linear Unit (ReLU) activation giving the best accuracy. Our model demonstrated superior performance compared to several existing approaches, including the method proposed by Chaunzwa et al.,[14] whose deep learning model, based on convolutional neural networks (CNNs) applied to CT images, achieved an AUC of 0.71. This highlights the potential advantages of radiomics features, which may provide a more interpretable and accessible approach for distinguishing NSCLC subtypes, particularly in scenarios with limited data. Unlike deep learning models, which require large datasets to capture complex feature representations effectively, radiomics features can extract meaningful information even when data is constrained. In comparison to Zhang et al.,[17] who achieved a good AUC of 0.870 with a combined model incorporating clinical-metabolic characteristics and radiomics, and Tang

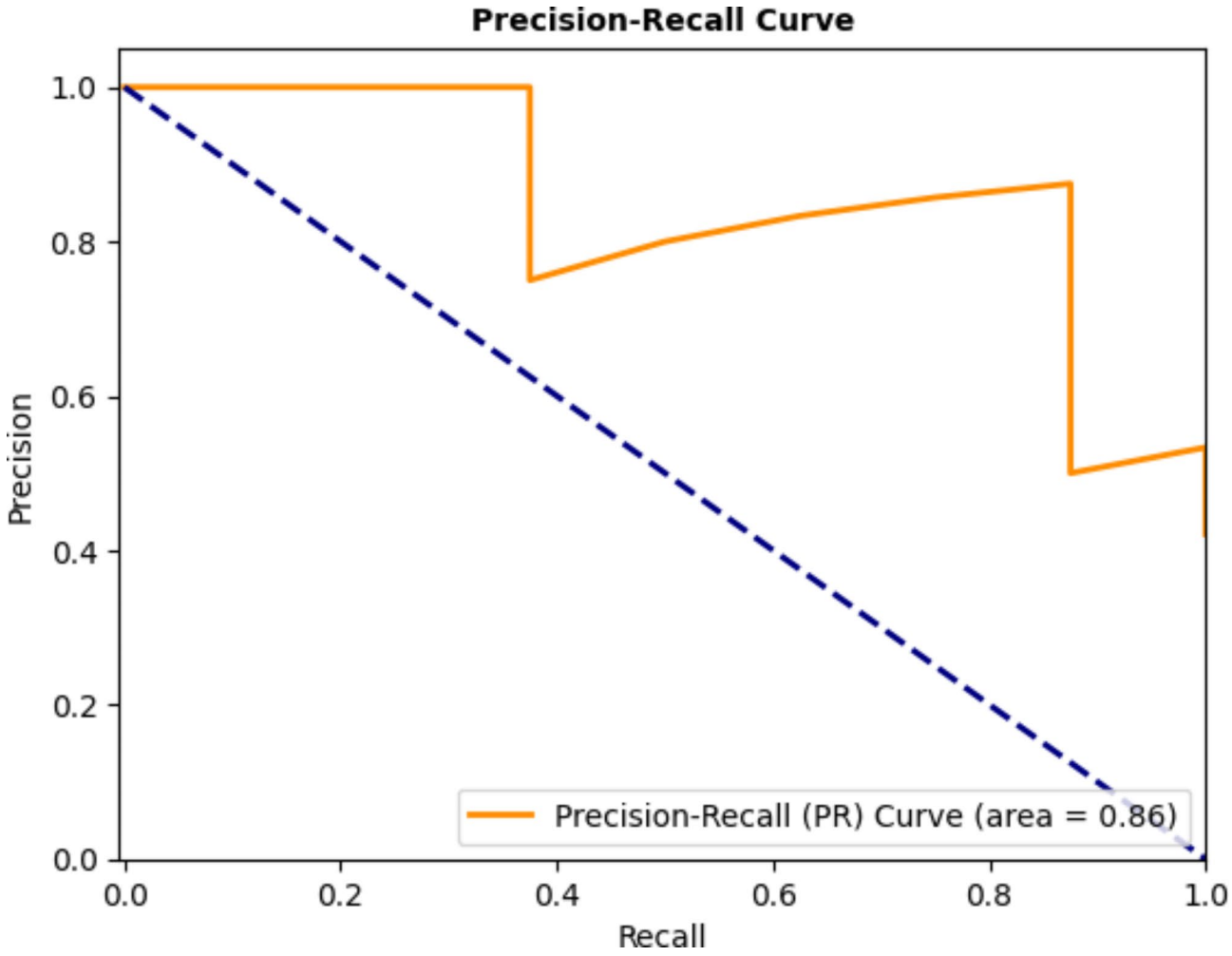


Fig. 7. Precision-Recall (PR) curve for the final classifier.

et al.,[18] who integrated PET and MRI features with an accuracy of 0.814, our model underscores the strength of a purely radiomics-based method without the need for additional clinical or multimodal imaging data. This makes our approach particularly significant as it highlights the potential of radiomics to achieve competitive accuracy with fewer data requirements, which may be crucial for clinical environments where advanced imaging modalities or comprehensive clinical data are not always readily available. Nonetheless, our findings suggest that future efforts integrating multimodal imaging and deep learning with radiomic features could further enhance predictive accuracy in distinguishing NSCLC subtypes.

In addition, decision curve analysis demonstrated the clinical utility of our predictive model, showing the highest net benefit between threshold probabilities of 0.0 and 0.3, and remaining clinically useful up to a threshold of 0.8. This highlights the model's ability to provide valuable decision-making support, particularly in identifying true positives at lower thresholds while minimising unnecessary treatments. However, we acknowledge that the study could benefit from further exploration with larger datasets, allowing for a broader range of machine learning models and more exhaustive hyperparameter tuning to identify the optimal approach. Although our power analysis, based on current sample sizes of 46 adenocarcinomas and 28 SCCs, indicates the study is well-powered (91%) to detect large effects, it shows lower power for detecting small or medium effects. Despite this limitation, our findings suggest that even with a modest sample size, it is possible to differentiate between AC and SCC nodules effectively and identify important radiomic features. This strengthens the clinical relevance of our results, supporting the use of radiomics-based approaches in distinguishing these tumour types. Future studies with larger datasets should further explore other machine learning algorithms to improve accuracy, stability, and clinical applicability in differentiating AC from SCC nodules.

These findings further underscore the potential of radiomic feature-based machine learning in improving diagnostic accuracy for lung nodule classification. By aiding radiologists in accurately distinguishing between SCC and adenocarcinoma, this approach holds promise for tailoring more effective treatment strategies, potentially leading to improved patient outcomes. Our previous radiomics-based models have similarly demonstrated their utility in distinguishing between benign and malignant pulmonary nodules, as well as differentiating COVID from other nodule types, which further supports the applicability of these methods in diverse clinical scenarios[20,21,32]. These outcomes emphasise the viability of radiomics as a complementary tool

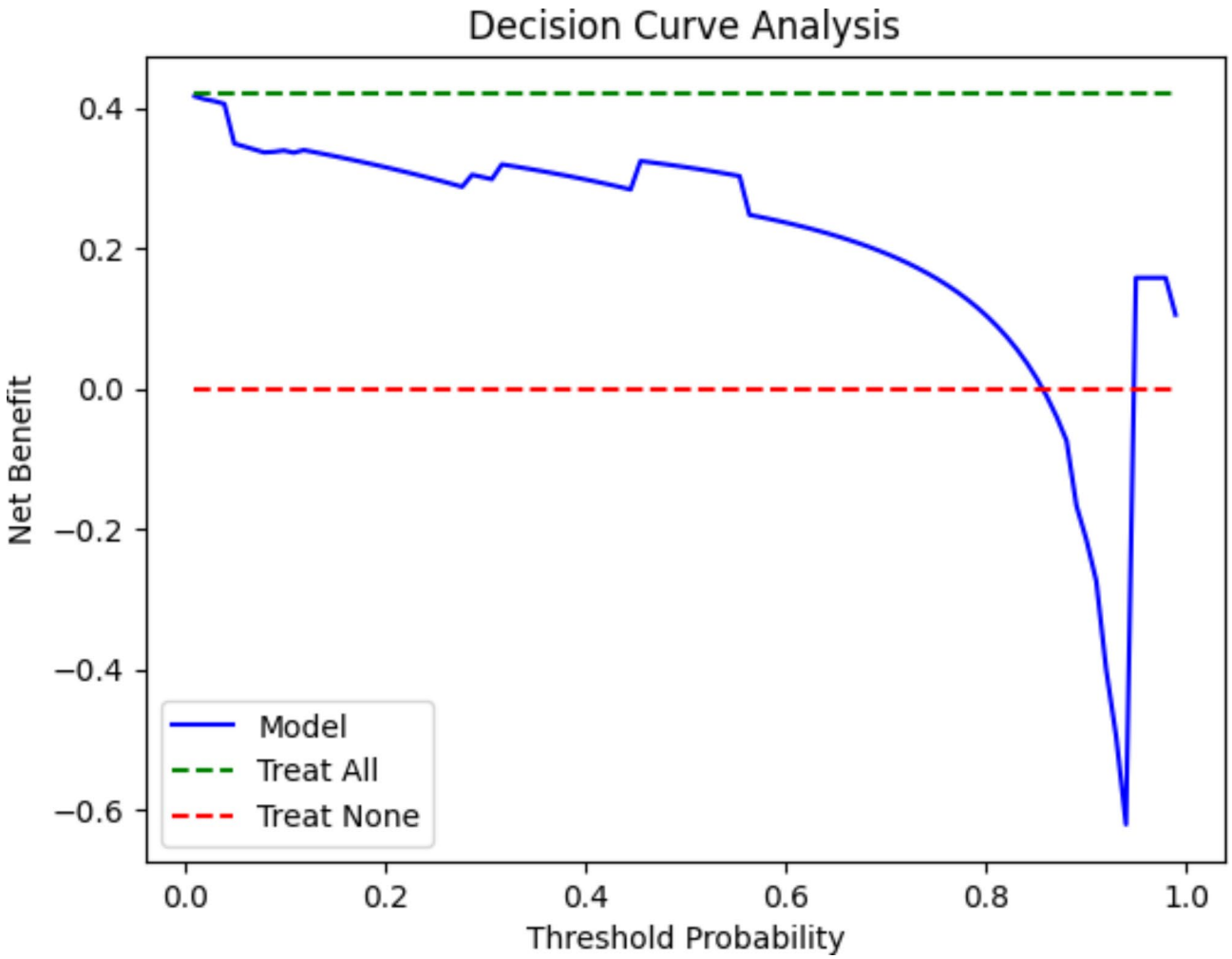


Fig. 8. Decision curve analysis for the final classifier.

in the differential diagnosis of NSCLC subtypes. However, realising its full potential requires further exploration through extensive multicenter trials, which are essential for validating its effectiveness across diverse patient cohorts and imaging platforms. Such collaborative efforts are pivotal for advancing the field and improving our understanding and management of lung nodules. Thus, our study highlights radiomics as an effective non-invasive adjunct to conventional CT imaging for distinguishing NSCLC subtypes. The integration of radiomics into clinical practice offers significant promise for enhancing diagnostic accuracy and patient management. Comprehensive multicenter trials will be crucial for validating its efficacy and facilitating adoption into clinical workflows.

Conclusion

In summary, our study highlights the clinical value of radiomics as a non-invasive tool for differentiating between NSCLC subtypes, particularly adenocarcinoma (AC) and squamous cell carcinoma (SCC). Accurate distinction between these subtypes is crucial for optimising treatment, as AC and SCC require different chemotherapy regimens and targeted therapies show varying efficacy. While histopathology remains the gold standard, it is limited by tumour heterogeneity and the risks associated with invasive biopsies. Radiomics offers a promising alternative by providing comprehensive, three-dimensional tumour analysis, improving diagnostic accuracy without the need for invasive procedures. Incorporating radiomics into routine clinical practice holds considerable promise for enhancing diagnostic interpretation, facilitating early diagnosis, assisting in treatment planning, and reducing the need for invasive biopsies. However, realising its full clinical utility hinges on executing comprehensive multicenter trials, which are essential for validating its efficacy and facilitating seamless integration into clinical workflows.

Supplement

The Supplement is available with this article.

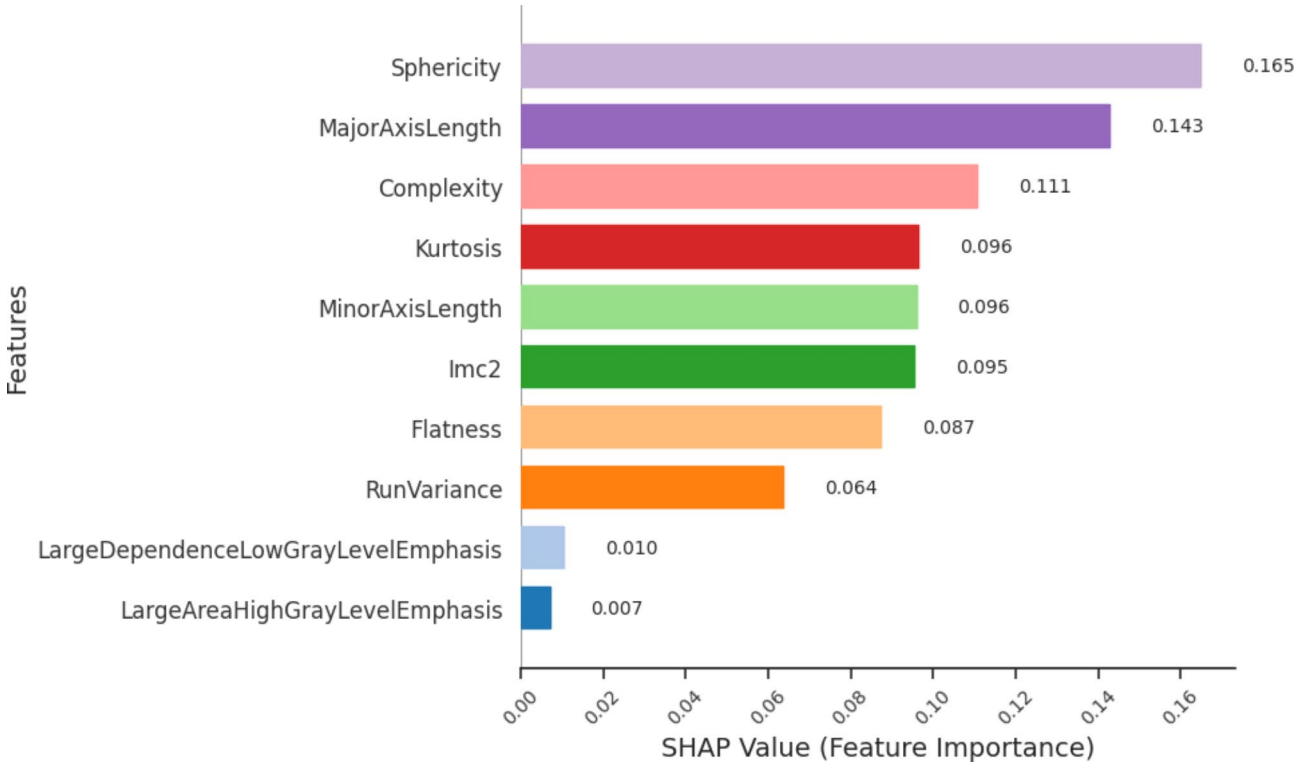


Fig. 9. Feature importance of using SHAP for the final best classifier.

Data availability

The datasets generated or analysed during the study are available from the corresponding author upon reasonable request.

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Author contributions

Conceptualization: MS, AC, AM. Data curation: MS. Formal analysis:: MS. Investigation: MS, AS. Methodology: MS, AS. Resources: MS, AS. Software: MS, AS, GK, SR. Supervision: AC, AM, GK. Writing—Original draft: MS, AS. Writing—Review & Editing: MS, AC.

Declarations

Competing interests

The authors declare no competing interests.

Ethical clearance

The study was performed after obtaining prior approval from the Institutional Research Ethics Committee – Sri Ramachandra Institute of Higher Education and Research (CSP–MED/19/SEP/56/122) and all methods were performed by relevant guidelines and regulations.

Informed consent

Informed consent was obtained from all subjects and/or their legal guardians involved in the study.

Additional information

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